

Chapter – 7 The Steady Magnetic Field

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
4	Formulate magnetic fields and its effects for a given distribution of moving charges using laws of magnetostatics.	Q3, A3, F	C5	2

Objectives:

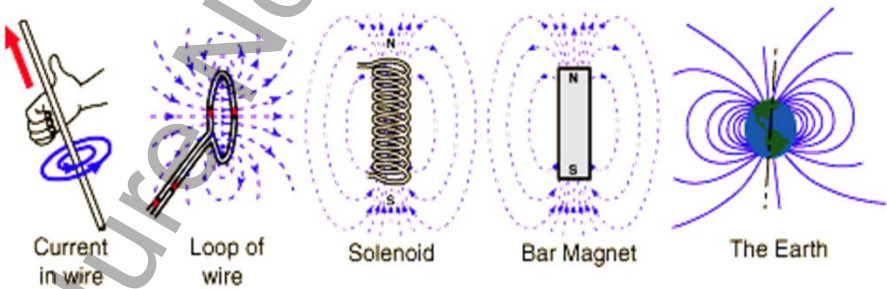
- 1. The Concept of Field (Source and Effect)
- 2. Magnetic Field Sources and Laws
 - a) Biot-Savart Law
 - b) Ampere’s Circuital Law
- 3. Curl (Another Vector Tool)
- 4. STOKES’ Theorem
- 5. Magnetic Flux and Density

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1

Magnetic Field Sources

Magnetic fields are produced by electric currents, which can be macroscopic currents in wires, or microscopic currents associated with electrons in atomic orbits.



Magnetic Field Sources

A time-varying Electric field is also a source of Magnetic Field

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Biot-Savart Law

At any point P, the magnitude of the magnetic field intensity produced by a differential element is proportional to the product of the current, the magnitude of the differential length, and the sine of the angle lying between the filament and a line connecting the filament to the point P at which the field is desired; also, the magnitude of the field is inversely proportional to the square of the distance from the filament to the point P. The constant of proportionality is $1/4\pi$

Free space

(Point 1)

$d\mathbf{L}_1$

$\mathbf{a}_{R12}$

$d\mathbf{H}$ (Inward)

(Point 2)

$d\mathbf{H}_2 = \frac{I_1 d\mathbf{L}_1 \times \mathbf{a}_{R12}}{4\pi R_{12}^2}$

$$d\mathbf{H} = \frac{I(d\mathbf{L} \times \mathbf{a}_R)}{4\pi R^2}$$

$$d\mathbf{H} = \frac{I(d\mathbf{L} \times \mathbf{R})}{4\pi R^3}$$

Biot-Savart = Ampere's law for the current element.

Magnetic Field Intensity, $\mathbf{H}$
Units: $\left[\frac{\text{A}}{\text{m}}\right]$

$$d\mathbf{E}_2 = \frac{dQ_1 \mathbf{a}_{R12}}{4\pi \epsilon_0 R_{12}^2}$$
$$\mathbf{H} = \oint \frac{I d\mathbf{L} \times \mathbf{a}_R}{4\pi R^2}$$

Verified Experimentally

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Biot-Savart Law

The direction of $d\mathbf{H}$ can be determined by the right-hand rule or right-handed screw rule.

Free space

$d\mathbf{H}$ (Inward)

(Point 1)

$d\mathbf{L}_1$

$\mathbf{a}_{R12}$

(Point 2)

$d\mathbf{H}_2 = \frac{I_1 d\mathbf{L}_1 \times \mathbf{a}_{R12}}{4\pi R_{12}^2}$

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4

Biot-Savart Law in terms of Distributed Sources

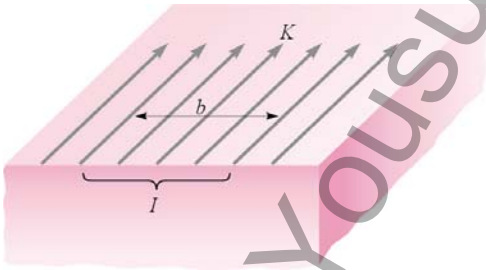
➤ The total current I within a transverse width b , in which there is a uniform surface current density K , is Kb .

$$I = Kb$$

where we assume that the width b is measured perpendicularly to the direction in which the current is flowing

➤ For a non-uniform surface current density, integration is necessary.

$$I = \int K \, dN$$



Surface Current Density, **K**
Units: A/m

Alternate Forms of Biot-Savart Law

$$IdL = KdS = Jdv$$

$$\mathbf{H} = \oint \frac{IdL \times \mathbf{a}_R}{4\pi R^2}$$

$$\mathbf{H} = \oint \frac{KdS \times \mathbf{a}_R}{4\pi R^2}$$

$$\mathbf{H} = \oint \frac{Jdv \times \mathbf{a}_R}{4\pi R^2}$$

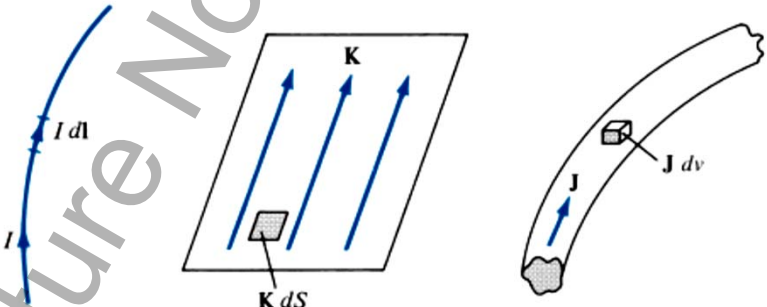
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5

Biot-Savart Law in terms of Distributed Sources

Alternate Forms of Biot-Savart Law

$$IdL = KdS = Jdv$$



$$\mathbf{H} = \oint \frac{IdL \times \mathbf{a}_R}{4\pi R^2}$$

$$\mathbf{H} = \oint \frac{KdS \times \mathbf{a}_R}{4\pi R^2}$$

$$\mathbf{H} = \oint \frac{Jdv \times \mathbf{a}_R}{4\pi R^2}$$

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Biot-Savart Law: Infinitely Long Straight Filament

To find Magnetic Field Intensity at Point 2.

SYMMETRY

1. With which coordinates does the field vary (or of what variables is H a function)?

2. Which components of $\mathbf{H}$ are present?

$\mathbf{H}(\mathbf{r}) = H_\rho \mathbf{a}_\rho + H_\phi \mathbf{a}_\phi + H_z \mathbf{a}_z$

$f(\rho, \phi, z)$

$\mathbf{H} = H_\phi(\rho) \mathbf{a}_\phi$

The direction of the magnetic field intensity vector is circumferential

CCS (ρ, ϕ, z)

Free space

$\mathbf{r} = \rho \mathbf{a}_\rho$
 $\mathbf{r}' = z' \mathbf{a}_z$

$\mathbf{R}_{12} = \rho \mathbf{a}_\rho - z' \mathbf{a}_z$

$d\mathbf{L}_1 = dz' \mathbf{a}_z$

$d\mathbf{H}_2 = \frac{I(d\mathbf{L}_1 \times \mathbf{a}_{R_{12}})}{4\pi R_{12}^2}$

$d\mathbf{H}_2 = \frac{I dz' \mathbf{a}_z \times (\rho \mathbf{a}_\rho - z' \mathbf{a}_z)}{4\pi(\rho^2 + z'^2)^{3/2}}$

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Biot-Savart Law: Infinitely Long Straight Filament

To find Magnetic Field Intensity at Point 2.

CCS (ρ, ϕ, z)

Electric current $\mathbf{I}$

$\mathbf{B}$ Magnetic field

$\mathbf{H}$ (or I) is out

$\mathbf{H}$ (or I) is in

cross section

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Biot-Savart Law: Infinitely Long Straight Filament

To find Magnetic Field Intensity at Point 2.

$$d\mathbf{H}_2 = \frac{I dz' \mathbf{a}_z \times (\rho \mathbf{a}_\rho - z' \mathbf{a}_z)}{4\pi(\rho^2 + z'^2)^{3/2}}$$

$$d\mathbf{H}_2 = \frac{I \rho dz' \mathbf{a}_\phi}{4\pi(\rho^2 + z'^2)^{3/2}}$$

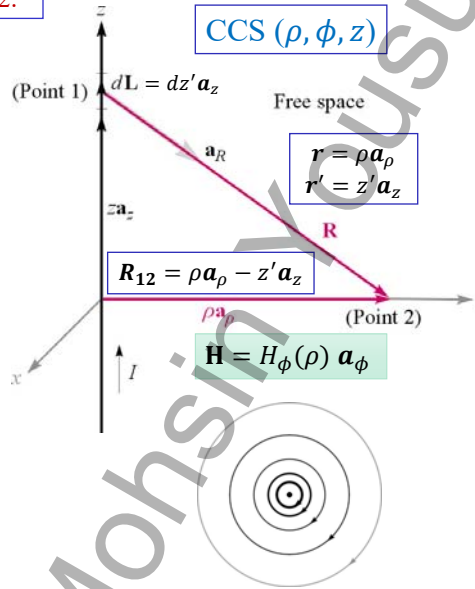
$$\mathbf{H}_2 = \int d\mathbf{H}_2 = \frac{I}{4\pi} \int_{-\infty}^{\infty} \frac{\rho dz' \mathbf{a}_\phi}{(\rho^2 + z'^2)^{3/2}}$$

$$\mathbf{a}_\phi = f(\phi), \quad \mathbf{a}_\phi \neq f(\rho \text{ or } z)$$

$$\mathbf{H}_2 = \frac{I \rho \mathbf{a}_\phi}{4\pi} \int_{-\infty}^{\infty} \frac{dz'}{(\rho^2 + z'^2)^{3/2}}$$

$$\mathbf{H}_2 = \frac{I \rho \mathbf{a}_\phi}{4\pi} \left(\frac{z'}{\rho^2 \sqrt{\rho^2 + z'^2}} \right)_{-\infty}^{\infty}$$

$$\mathbf{H}_2 = \frac{I}{2\pi \rho} \mathbf{a}_\phi$$



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9

Biot-Savart Law: Finite Straight Filament

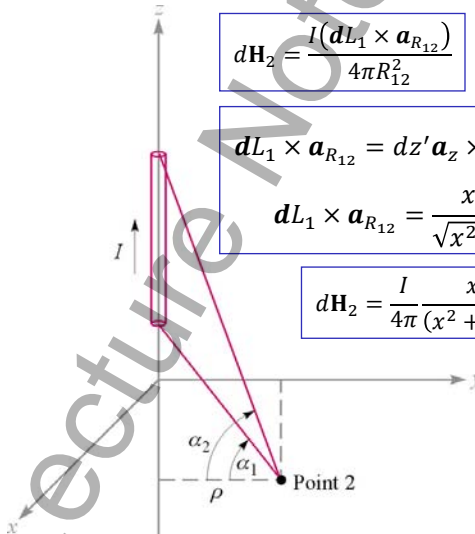
To find Magnetic Field Intensity at Point 2.

$$d\mathbf{H}_2 = \frac{I(d\mathbf{L}_1 \times \mathbf{a}_{R_{12}})}{4\pi R_{12}^2}$$

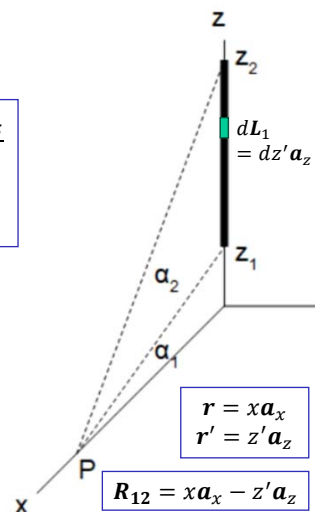
$$d\mathbf{L}_1 \times \mathbf{a}_{R_{12}} = dz' \mathbf{a}_z \times \frac{x \mathbf{a}_x - z' \mathbf{a}_z}{\sqrt{x^2 + z'^2}}$$

$$d\mathbf{L}_1 \times \mathbf{a}_{R_{12}} = \frac{x dz'}{\sqrt{x^2 + z'^2}} \mathbf{a}_y$$

$$d\mathbf{H}_2 = \frac{I}{4\pi} \frac{x dz'}{(x^2 + z'^2)^{3/2}} \mathbf{a}_y$$



RCS (x, y, z)



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Biot-Savart Law: Finite Straight Filament

To find Magnetic Field Intensity at Point 2.

RCS (x, y, z)

$$\mathbf{H}_2 = \frac{Ix}{4\pi} \mathbf{a}_y \int_{z_1}^{z_2} \frac{dz'}{(x^2 + z'^2)^{3/2}}$$
$$\mathbf{H}_2 = \frac{Ix}{4\pi} \mathbf{a}_y \left(\frac{z'}{x^2 \sqrt{x^2 + z'^2}} \right)_{z_1}^{z_2}$$
$$\mathbf{H}_2 = \frac{I}{4\pi x} \mathbf{a}_y \left[\frac{z_2}{\sqrt{x^2 + z_2^2}} - \frac{z_1}{\sqrt{x^2 + z_1^2}} \right]$$
$$\mathbf{H}_2 = \frac{I}{4\pi x} (\sin \alpha_2 - \sin \alpha_1) \mathbf{a}_y$$

The problem is symmetric w.r.t. rotations about the z-axis, $x \rightarrow \rho, \mathbf{a}_y \rightarrow \mathbf{a}_\phi$

$$\mathbf{H}_2 = \frac{I}{4\pi \rho} (\sin \alpha_2 - \sin \alpha_1) \mathbf{a}_\phi$$

$$\mathbf{r} = x\mathbf{a}_x$$
$$\mathbf{r}' = z'\mathbf{a}_z$$
$$\mathbf{R}_{12} = x\mathbf{a}_x - z'\mathbf{a}_z$$

If one or both ends are below point 2, then α_1 is or both α_1 and α_2 are negative.

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Example 7.1

Determine $\mathbf{H}$ at $P(0.4, 0.3, 0)$ in the field of an 8 A filamentary current is directed inward from infinity to the origin on the positive x-axis, and then outward to infinity along the y-axis

$$\mathbf{H}_2 = \frac{I}{4\pi \rho} (\sin \alpha_2 - \sin \alpha_1) \mathbf{a}_\phi$$

Semi-infinite current on the x axis, $\rho_x = 0.3$

$$\mathbf{H}_{2(x)} = \frac{8}{4\pi(0.3)} (\sin \alpha_{2x} - \sin \alpha_{1x}) \mathbf{a}_\phi$$
$$\mathbf{H}_{2(x)} = \frac{8}{4\pi(0.3)} (\sin 53.1^\circ + 1) \mathbf{a}_\phi$$
$$\mathbf{H}_{2(x)} = \frac{2}{\pi(0.3)} (1.8) \mathbf{a}_\phi = \frac{12}{\pi} \mathbf{a}_\phi$$

The unit vector $\mathbf{a}_\phi$ becomes $-\mathbf{a}_z$

$$\mathbf{H}_{2(x)} = -\frac{12}{\pi} \mathbf{a}_z, \quad [\text{A/m}]$$

$$\alpha_{1x} = -90^\circ$$
$$\alpha_{2x} = \tan^{-1} \frac{0.4}{0.3} = 53.1^\circ$$
$$\alpha_{2y} = 90^\circ$$
$$\alpha_{1y} = -\tan^{-1} \frac{0.3}{0.4} = -36.9^\circ$$

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Example 7.1

Determine **H** at $P(0.4, 0.3, 0)$ in the field of an 8 A filamentary current is directed inward from infinity to the origin on the positive x-axis, and then outward to infinity along the y-axis

$$\mathbf{H}_2 = \frac{I}{4\pi\rho} (\sin \alpha_2 - \sin \alpha_1) \mathbf{a}_\phi$$

Semi-infinite current on the y axis, $\rho_y = 0.4$

$$\mathbf{H}_{2(y)} = \frac{8}{4\pi(0.4)} (\sin \alpha_{2y} - \sin \alpha_{1y}) \mathbf{a}_\phi$$
$$\mathbf{H}_{2(y)} = \frac{8}{4\pi(0.4)} (1 + \sin 36.9^\circ) \mathbf{a}_\phi$$
$$\mathbf{H}_{2(y)} = \frac{2}{\pi(0.4)} (1.64) \mathbf{a}_\phi = \frac{8}{\pi} \mathbf{a}_\phi$$

The unit vector $\mathbf{a}_\phi$ becomes $-\mathbf{a}_z$

$$\mathbf{H}_{2(x)} = -\frac{12}{\pi} \mathbf{a}_z, \quad [A/m]$$

$$\mathbf{H}_{2(y)} = -\frac{8}{\pi} \mathbf{a}_z, \quad [A/m]$$

$$\mathbf{H}_2 = \mathbf{H}_{2(x)} + \mathbf{H}_{2(y)} = -\frac{20}{\pi} \mathbf{a}_z = -6.37 \mathbf{a}_z \quad [A/m]$$

$\alpha_{1x} = -90^\circ$
 $\alpha_{2x} = \tan^{-1} \frac{0.4}{0.3} = 53.1^\circ$
 $\alpha_{2y} = 90^\circ$
 $\alpha_{1y} = -\tan^{-1} \frac{0.3}{0.4} = -36.9^\circ$

D7.1. D7.2.

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Ampere's Circital Law or Ampere's Work Law

Statement:

Ampere's Circital Law states that *the line integral of **H** about any closed path is exactly equal to the direct current enclosed by the path.*

$$\oint \mathbf{H} \cdot d\mathbf{L} = I_{encl}$$

We define positive current as flowing in the direction of the advance of a right-handed screw turned in the direction in which the closed path is traversed.

The magnetic field in space around an electric current is proportional to the electric current which serves as its source, just as the electric field in space is proportional to the charge which serves as its source.

A conductor has a total current I . The line integral of $\mathbf{H}$ about the closed paths a and b is equal to I , and the integral around path c is less than I , since the entire current is not enclosed by the path.

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Ampere's Circital Law: Infinitely Long Straight Filament

To find Magnetic Field Intensity at Point 2.

SYMMETRY IS MUST

1. With which coordinates does the field vary (or of what variables is H a function)?
2. Which components of $\mathbf{H}$ are present?

$\mathbf{H}(\mathbf{r}) = H_\rho \mathbf{a}_\rho + H_\phi \mathbf{a}_\phi + H_z \mathbf{a}_z$

$f(\rho, \phi, z)$

$\mathbf{H} = H_\phi(\rho) \mathbf{a}_\phi$

The direction of the magnetic field intensity vector is *circumferential*

CCS (ρ, ϕ, z)

Free space

(Point 1)

$d\mathbf{L} = dz' \mathbf{a}_z$

$\mathbf{a}_R$

$\mathbf{a}_z$

$\mathbf{R}$

(Point 2)

$\rho \mathbf{a}_\rho$

$\oint \mathbf{H} \cdot d\mathbf{L} = I_{encl}$

A path is chosen for which $\mathbf{H}$ is either perpendicular or tangential, and along which H is constant.

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Ampere's Circital Law: Infinitely Long Straight Filament

To find Magnetic Field Intensity at Point 2.

The path must be a circle of radius ρ , and Ampere's Circital Law becomes,

$\oint \mathbf{H} \cdot d\mathbf{L} = \int_\phi H_\phi \mathbf{a}_\phi \cdot \rho d\phi \mathbf{a}_\phi$

$\oint \mathbf{H} \cdot d\mathbf{L} = \int_{\phi=0}^{2\pi} H_\phi \rho d\phi$

$\oint \mathbf{H} \cdot d\mathbf{L} = H_\phi \rho \int_{\phi=0}^{2\pi} d\phi = H_\phi 2\pi \rho = I$

$H_\phi = \frac{I}{2\pi \rho}$

$\mathbf{H} = \frac{I}{2\pi \rho} \mathbf{a}_\phi$

CCS (ρ, ϕ, z)

Free space

(Point 1)

$d\mathbf{L}$

$\mathbf{a}_R$

$\mathbf{a}_z$

$\mathbf{H} = H_\phi(\rho) \mathbf{a}_\phi$

$d\mathbf{L}_\phi = \rho d\phi \mathbf{a}_\phi$

$\rho \mathbf{a}_\rho$

(Point 2)

I

Ampere's Circital Law becomes,

$\oint \mathbf{H} \cdot d\mathbf{L} = \int_\phi H_\phi \mathbf{a}_\phi \cdot \rho d\phi \mathbf{a}_\phi$

$\oint \mathbf{H} \cdot d\mathbf{L} = \int_{\phi=0}^{2\pi} H_\phi \rho d\phi$

$\oint \mathbf{H} \cdot d\mathbf{L} = H_\phi \rho \int_{\phi=0}^{2\pi} d\phi = H_\phi 2\pi \rho = I$

$H_\phi = \frac{I}{2\pi \rho}$

$\mathbf{H} = \frac{I}{2\pi \rho} \mathbf{a}_\phi$

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Ampere's Circuital Law: Infinitely Long Coaxial Cable

To find **H** everywhere in the transmission line carries a uniformly distributed total current I in the center conductor and $-I$ in the outer conductor.

SYMMETRY IS MUST

$$\mathbf{H}(\mathbf{r}) = H_\rho \mathbf{a}_\rho + H_\phi \mathbf{a}_\phi + H_z \mathbf{a}_z$$

$f(\rho, \phi, z)$

$$\mathbf{H} = H_\phi(\rho) \mathbf{a}_\phi$$

CCS (ρ, ϕ, z)

$$\oint \mathbf{H} \cdot d\mathbf{L} = I_{encl}$$

Top View

$$H = \begin{cases} H_\phi, & \rho < a \\ H_\phi, & a < \rho < b \\ H_\phi, & b < \rho < c \\ H_\phi, & \rho > c \end{cases}$$

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17

Ampere's Circuital Law: Infinitely Long Coaxial Cable

To find **H** everywhere in the transmission line carries a uniformly distributed total current I in the center conductor and $-I$ in the outer conductor.

CCS (ρ, ϕ, z)

$$\mathbf{H} = H_\phi(\rho) \mathbf{a}_\phi$$
$$\oint \mathbf{H} \cdot d\mathbf{L} = I_{encl}$$

CASE 1: $a < \rho < b$

$$\begin{aligned} LHS &= 2\pi\rho H_\phi \\ RHS &= I_{encl} = I \\ H_\phi &= \frac{I}{2\pi\rho} \end{aligned}$$

CASE 2: $\rho < a$

$$\begin{aligned} RHS &= I_{encl} = I \frac{\pi\rho^2}{\pi a^2} = I \frac{\rho^2}{a^2} \\ H_\phi &= I \frac{\rho}{2\pi a^2} \end{aligned}$$

CASE 3: $\rho > c$

$$\begin{aligned} RHS &= I_{encl} = I - I = 0 \\ H_\phi &= 0 \end{aligned}$$
$$H = \begin{cases} H_\phi, & \rho < a \\ H_\phi, & a < \rho < b \\ H_\phi, & b < \rho < c \\ H_\phi, & \rho > c \end{cases}$$

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18

Ampere's Circuital Law: Infinitely Long Coaxial Cable

To find **H** everywhere in the transmission line carries a uniformly distributed total current I in the center conductor and $-I$ in the outer conductor.

CASE 4: $b < \rho < c$

$CCS(\rho, \phi, z)$

$\mathbf{H} = H_{\phi}(\rho) \mathbf{a}_{\phi}$

$\oint \mathbf{H} \cdot d\mathbf{L} = I_{encl}$

$$H_{\phi} = \begin{cases} \frac{I \rho}{2\pi a^2}, & \rho < a \\ \frac{I}{2\pi \rho}, & a < \rho < b \\ \frac{I}{2\pi \rho} \frac{c^2 - \rho^2}{c^2 - b^2}, & b < \rho < c \\ 0, & \rho > c \end{cases}$$

$LHS = 2\pi \rho H_{\phi}$

$RHS = I_{encl} = I - I \frac{\rho^2 - b^2}{c^2 - b^2}$

$I_{encl} = I \frac{c^2 - \rho^2}{c^2 - b^2}$

$H_{\phi} = \frac{I}{2\pi \rho} \frac{c^2 - \rho^2}{c^2 - b^2}$

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Ampere's Circuital Law: Infinite Sheet of Current

To find **H** above and below a sheet of current flowing in the positive y direction and located in the $z = 0$ plane.

SYMMETRY IS MUST

$\mathbf{H}(\mathbf{r}) = H_x \mathbf{a}_x + H_y \mathbf{a}_y + H_z \mathbf{a}_z$

$f(x, y, z)$

$\mathbf{H} = H_x(z) \mathbf{a}_x$

$RCS(x, y, z)$

$\oint \mathbf{H} \cdot d\mathbf{L} = I_{encl}$

A path 1 – 1' – 2' – 2 – 1 composed of straight-line segments that are either parallel or perpendicular to H_x is chosen.

$H_{x1}L + H_{x2}(-L) = K_y L$

$H_{x1} - H_{x2} = K_y$

path 3 – 3' – 2' – 2 – 3

$H_{x3} - H_{x2} = K_y$ $H_{x3} = H_{x1}$

It follows that H_x is the same for all positive z . Similarly, H_x is the same for all negative z .

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Ampere's Circital Law: Infinite Sheet of Current

To find **H** above and below a sheet of current flowing in the positive *y* direction and located in the *z* = 0 plane.

Because of the *symmetry*, then, the magnetic field intensity on one side of the current sheet is the negative of that on the other.

$$H_x = \frac{1}{2} K_y, \quad z > 0$$
$$H_x = -\frac{1}{2} K_y, \quad z < 0$$

Letting **a_N** be a unit vector normal (outward) to the current sheet, the result may be written in a form correct for all *z* as,

$$\mathbf{H} = \frac{1}{2} \mathbf{K} \times \mathbf{a}_N, \quad (1)$$

If a second sheet of current is placed with, *K* = −*K_y**a_y*, is placed at *z* = *h*, eq(1) implies,

$$\mathbf{H} = \mathbf{K} \times \mathbf{a}_N, \quad (0 < z < h)$$
$$\mathbf{H} = 0, \quad (z < 0, z > h)$$

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Practice Problem

Planes *z* = 0 and *z* = 4 carry a surface current **K** = −10 **a_x** $\frac{\text{A}}{\text{m}}$ and **K** = 10 **a_x** $\frac{\text{A}}{\text{m}}$, respectively. Determine **H** at

(a) (1,1,1)
(b) (0,−3,10)

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Ampere's Circuital Law: Infinitely Long Solenoid

To find $\mathbf{H}$ due to an infinitely long solenoid of radius a and uniform current density $K_a \mathbf{a}_\phi$.

CCS (ρ, ϕ, z)

$\oint \mathbf{H} \cdot d\mathbf{L} = I_{encl}$

A path is chosen for which $\mathbf{H}$ is either perpendicular or tangential, and along which H is constant.

$\mathbf{H}(\mathbf{r}) = H_\rho \mathbf{a}_\rho + H_\phi \mathbf{a}_\phi + H_z \mathbf{a}_z$

SYMMETRY IS MUST

$f(\rho, \phi, z)$

$\mathbf{H} = H_z(\rho) \mathbf{a}_z$

$RHS = I_{encl} = K_a d$
 $LHS = \oint \mathbf{H} \cdot d\mathbf{L} = H_z d$
 $H_z = K_a$

If there is an additional hollow concentric cylinder inside the original one with radius $b < a$ and current $\mathbf{K} = K_b \mathbf{a}_\phi$, we calculate the field $\mathbf{H}$ everywhere.

- The field will be zero outside the outer cylinder.
- Inside the inner cylinder, both cylinders contribute in the same way, so the field is

$H_z = K_a + K_b, \quad \rho < b$ $H_z = K_a, \quad b < \rho < a$

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Ampere's Circuital Law: Infinitely Long Solenoid

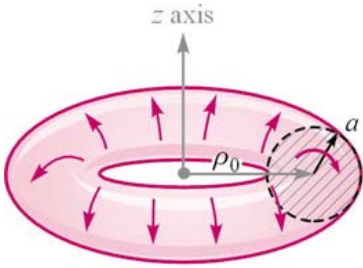
To find $\mathbf{H}$ due to an infinitely long solenoid of radius a and uniform current density $K_a \mathbf{a}_\phi$.

$\oint \mathbf{H} \cdot d\mathbf{L} = I_{encl}$

$\mathbf{H} = \frac{NI}{d} \mathbf{a}_z$
(well inside coil)

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Ampere's Circuital Law – Example of Toroid

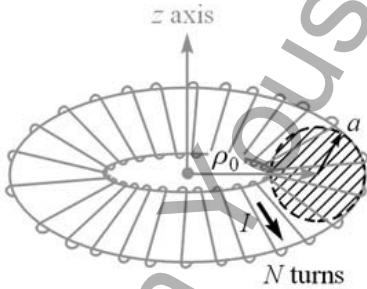


$\mathbf{K} = K_a \mathbf{a}_z$ at $\rho = \rho_0 - a, z = 0$

$\mathbf{H} = K_a \frac{\rho_0 - a}{\rho} \mathbf{a}_\phi$ (inside toroid)

$\mathbf{H} = 0$ (outside)

(a)



$\mathbf{H} = \frac{NI}{2\pi\rho} \mathbf{a}_\phi$ (well inside toroid)

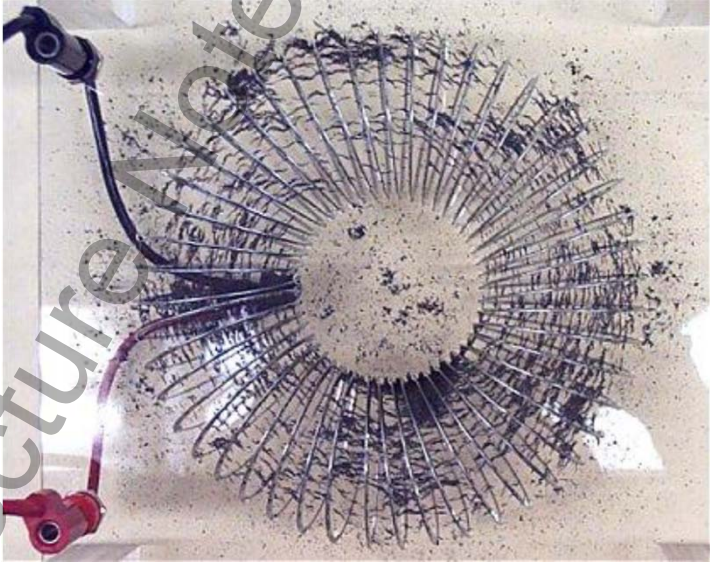
(b)

D7.3.

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25

Ampere's Circuital Law – Field of a Toroid

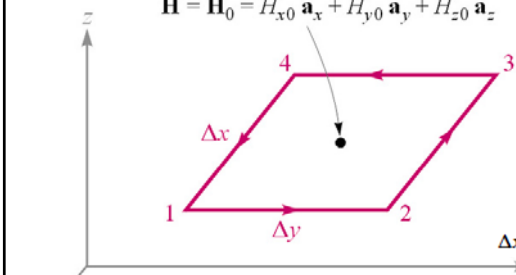


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26

CURL

The curl of a vector function is the vector product of the del operator with a vector function

$$\mathbf{H} = \mathbf{H}_0 = H_{x0} \mathbf{a}_x + H_{y0} \mathbf{a}_y + H_{z0} \mathbf{a}_z$$


$$\text{curl } \mathbf{H} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix}$$
$$\text{curl } \mathbf{H} = \nabla \times \mathbf{H}$$

$$(\text{curl } \mathbf{H})_N = \lim_{\Delta S_N \rightarrow 0} \frac{\oint \mathbf{H} \cdot d\mathbf{L}}{\Delta S_N}$$
$$\lim_{\Delta x, \Delta y \rightarrow 0} \frac{\oint \mathbf{H} \cdot d\mathbf{L}}{\Delta x \Delta y} = \frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = J_z$$
$$\lim_{\Delta y, \Delta z \rightarrow 0} \frac{\oint \mathbf{H} \cdot d\mathbf{L}}{\Delta y \Delta z} = \frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = J_x$$
$$\lim_{\Delta z, \Delta x \rightarrow 0} \frac{\oint \mathbf{H} \cdot d\mathbf{L}}{\Delta z \Delta x} = \frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} = J_y$$
$$\text{curl } \mathbf{H} = \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) \mathbf{a}_x + \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) \mathbf{a}_y + \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \mathbf{a}_z$$

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27

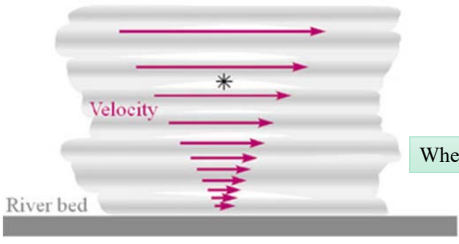
CURL: Physical Interpretation & Other Coordinate Systems

$$\nabla \times \mathbf{H} = \left(\frac{1}{\rho} \frac{\partial H_z}{\partial \phi} - \frac{\partial H_\phi}{\partial z} \right) \mathbf{a}_\rho + \left(\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial \rho} \right) \mathbf{a}_\phi + \left(\frac{1}{\rho} \frac{\partial (\rho H_\phi)}{\partial \rho} - \frac{1}{\rho} \frac{\partial H_\rho}{\partial \phi} \right) \mathbf{a}_z \quad (\text{cylindrical})$$
$$\nabla \times \mathbf{H} = \frac{1}{r \sin \theta} \left(\frac{\partial (H_\phi \sin \theta)}{\partial \theta} - \frac{\partial H_\theta}{\partial \phi} \right) \mathbf{a}_r + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial H_r}{\partial \phi} - \frac{\partial (r H_\phi)}{\partial r} \right) \mathbf{a}_\theta + \frac{1}{r} \left(\frac{\partial (r H_\theta)}{\partial r} - \frac{\partial H_r}{\partial \theta} \right) \mathbf{a}_\phi \quad (\text{spherical})$$

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CURL – Circulation per unit area

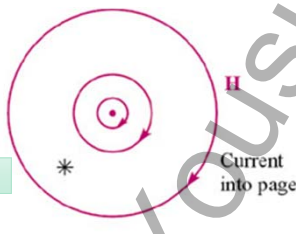


Velocity

River bed

Wheel Paddle

(a)



Current into page

(b)

Curl of $\mathbf{E}$ must be zero, for the circulation is zero.

$\nabla \times \mathbf{E} = 0$

3rd Maxwell's Equation

Curl of $\mathbf{H}$ is not zero, for the circulation is equal to current density by Ampere's circuital law.

$\nabla \times \mathbf{H} = \mathbf{J}$

2nd Maxwell's Equation

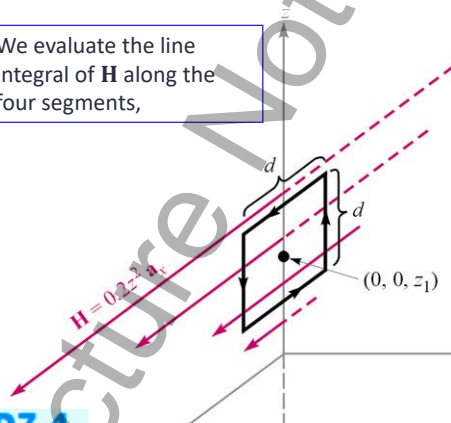
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Example 7.2

Suppose that $\mathbf{H} = 0.2z^2 \mathbf{a}_x$ for $z > 0$, and $\mathbf{H} = 0$ elsewhere, Calculate $\oint \mathbf{H} \cdot d\mathbf{L}$ about a square path with side d , centered at $(0, 0, z_1)$ in the $y = 0$ plane where $z_1 > d/2$.

We evaluate the line integral of $\mathbf{H}$ along the four segments,



$\mathbf{H} = 0.2z^2 \mathbf{a}_x$

$\mathbf{H} = 0$

$(0, 0, z_1)$

d

d

$$\oint \mathbf{H} \cdot d\mathbf{L} = 0.2 \left(z_1 + \frac{d}{2} \right)^2 d + 0 - 0.2 \left(z_1 - \frac{d}{2} \right)^2 d + 0$$
$$\oint \mathbf{H} \cdot d\mathbf{L} = 0.4z_1 d^2$$

In the limit as the area approaches zero, we find

$$(\nabla \times \mathbf{H})_y = \lim_{d \rightarrow 0} \frac{\oint \mathbf{H} \cdot d\mathbf{L}}{d^2} = \lim_{d \rightarrow 0} \frac{0.4z_1 d^2}{d^2} = 0.4z_1$$

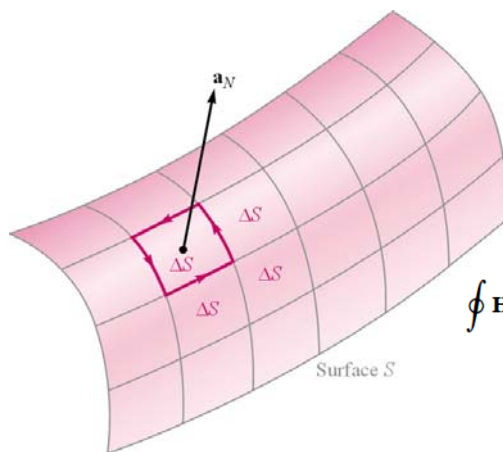
$$\nabla \times \mathbf{H} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0.2z^2 & 0 & 0 \end{vmatrix} = \frac{\partial}{\partial z} (0.2z^2) \mathbf{a}_y = 0.4z \mathbf{a}_y$$

$\nabla \times \mathbf{H} = 0.4z_1 \mathbf{a}_y$

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15

Stokes' Theorem



$$\frac{\oint \mathbf{H} \cdot d\mathbf{L}_{\Delta S}}{\Delta S} \doteq (\nabla \times \mathbf{H})_N$$

$$\frac{\oint \mathbf{H} \cdot d\mathbf{L}_{\Delta S}}{\Delta S} \doteq (\nabla \times \mathbf{H}) \cdot \mathbf{a}_N$$

$$\oint \mathbf{H} \cdot d\mathbf{L}_{\Delta S} \doteq (\nabla \times \mathbf{H}) \cdot \mathbf{a}_N \Delta S = (\nabla \times \mathbf{H}) \cdot \Delta \mathbf{S}$$

$$\oint \mathbf{H} \cdot d\mathbf{L} \equiv \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S}$$

The sum of the closed line integrals about the perimeter of every ΔS is the same as the closed line integral about the perimeter of S because of cancellation on every path.

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31

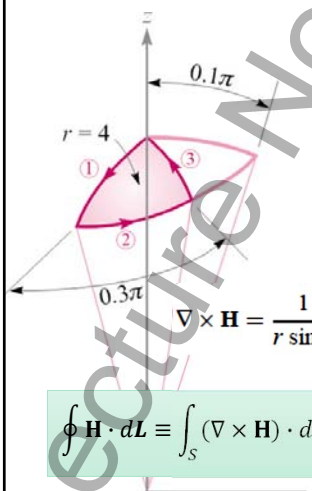
Example 7.3 $\mathbf{H} = 6r \sin \phi \mathbf{a}_r + 18r \sin \theta \cos \phi \mathbf{a}_\phi$

Solution. The first path segment is described in spherical coordinates by $r = 4$, $0 \leq \theta \leq 0.1\pi$, $\phi = 0$; the second one by $r = 4$, $\theta = 0.1\pi$, $0 \leq \phi \leq 0.3\pi$; and the third by $r = 4$, $0 \leq \theta \leq 0.1\pi$, $\phi = 0.3\pi$. The differential path element $d\mathbf{L}$ is the vector

$$d\mathbf{L} = dr \mathbf{a}_r + r d\theta \mathbf{a}_\theta + r \sin \theta d\phi \mathbf{a}_\phi$$

$$\oint \mathbf{H} \cdot d\mathbf{L} = \int_1 H_\theta r d\theta + \int_2 H_\phi r \sin \theta d\phi + \int_3 H_\theta r d\theta$$

$$\begin{aligned} \oint \mathbf{H} \cdot d\mathbf{L} &= \int_0^{0.3\pi} [18(4) \sin 0.1\pi \cos \phi] 4 \sin 0.1\pi d\phi \\ &= 288 \sin^2 0.1\pi \sin 0.3\pi = 22.2 \text{ A} \end{aligned}$$



$$\nabla \times \mathbf{H} = \frac{1}{r \sin \theta} (36r \sin \theta \cos \theta \cos \phi) \mathbf{a}_r + \frac{1}{r} \left(\frac{1}{\sin \theta} 6r \cos \phi - 36r \sin \theta \cos \phi \right) \mathbf{a}_\theta$$

$$\begin{aligned} \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} &= \int_0^{0.3\pi} \int_0^{0.1\pi} (36 \cos \theta \cos \phi) 16 \sin \theta d\theta d\phi \\ &= \int_0^{0.3\pi} 576 \left(\frac{1}{2} \sin^2 \theta \right) \Big|_0^{0.1\pi} \cos \phi d\phi \\ &= 288 \sin^2 0.1\pi \sin 0.3\pi = 22.2 \text{ A} \end{aligned}$$

$$\oint \mathbf{H} \cdot d\mathbf{L} \equiv \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S}$$

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32

Magnetic Flux and Magnetic Flux Density

$$\mathbf{B} = \mu_0 \mathbf{H} \qquad \mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

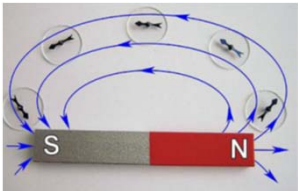
Units of B: Wb/m² or Tesla (T)
 μ_0 is the permeability of free space

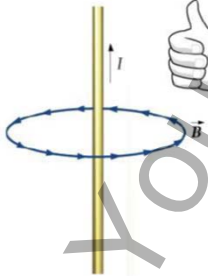
$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{S}$$

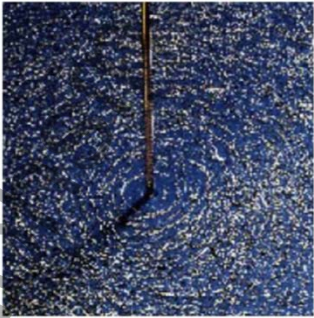
$$\Psi = \oint_S \mathbf{D} \cdot d\mathbf{S}$$

➤ Magnetic flux line is a path to which **B** is tangential at every point on the line.

➤ Each flux line is **closed** and has no beginning or end.







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Magnetic Flux Lines of a Broken Magnet

It is not possible to isolate the north and south poles of a magnet

N	N	N	N
		S	S
		N	N
		S	S
	N	N	N
		S	S
		N	N
		S	S
S	S	S	S

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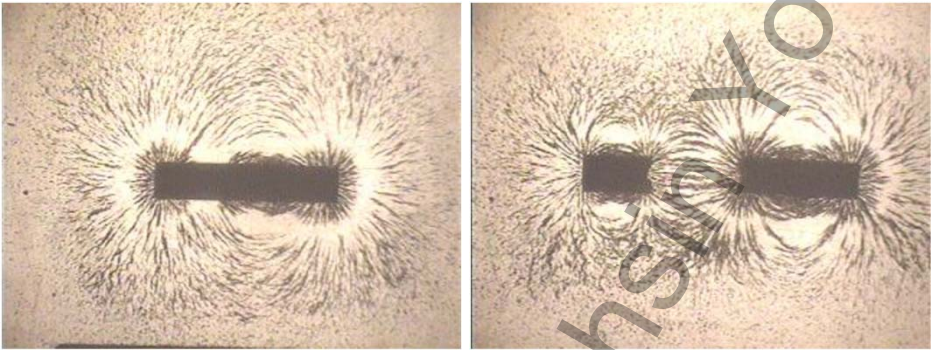
34

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17

Magnetic Flux Lines of a Broken Magnet

It is not possible to isolate the north and south poles of a magnet

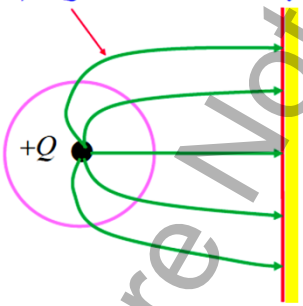


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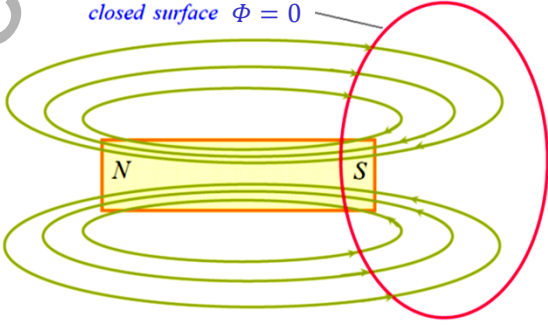
35

Magnetic Flux and Magnetic Flux Density

$\psi = Q$ over the closed surface



closed surface $\Phi = 0$



➤ In an electrostatic field the flux crossing a closed surface is the same as the charge enclosed.

➤ $\Psi = \oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{encl}$

➤ So it is possible to have an isolated electric charge and the flux lines produced by it need not be closed.

➤ Unlike electric flux lines, magnetic flux lines always close upon themselves.

➤ This is because it is not possible to have isolated magnetic poles (or magnetic charges).

➤ $\Phi = \oint_S \mathbf{B} \cdot d\mathbf{S} = 0$

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36

Gauss's Law for Magnetostatic Fields

The total flux through a closed surface in a magnetic field is zero

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{S}$$

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

Applying Divergence Theorem,

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{vol} \nabla \cdot \mathbf{D} dv$$

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = \int_{vol} \nabla \cdot \mathbf{B} dv = 0$$

$$\nabla \cdot \mathbf{B} = 0$$

4th Maxwell's Equation

It states that magnetostatic fields have no **sources or sinks**.

D7.7.

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Magnetostatic Fields: Example of Coaxial Line

Let us find the flux between the conductors of the coaxial line for $a < \rho < b$.

$$H = \frac{I}{2\pi\rho}, \quad (a < \rho < b)$$

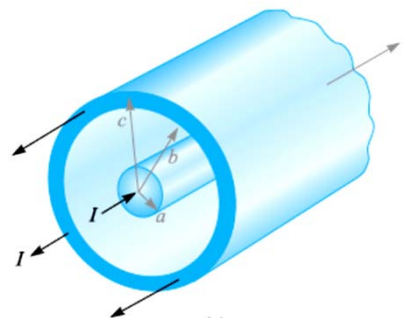
$$\mathbf{B} = \mu_0 \mathbf{H} = \frac{\mu_0 I}{2\pi\rho} \mathbf{a}_\phi$$

The magnetic flux contained between the conductors in a length d is the flux crossing any radial plane extending from $\rho = a$ to $\rho = b$ and from, say, $z = 0$ to $z = d$

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{S} = \int_{z=0}^d \int_{\rho=a}^b \frac{\mu_0 I}{2\pi\rho} \mathbf{a}_\phi \cdot d\rho dz \mathbf{a}_\phi$$

$$\Phi = \frac{\mu_0 I d}{2\pi} \ln \frac{b}{a}$$

This expression will be used later to obtain the **inductance** of the coaxial transmission line.



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Maxwell's Equations for Static Fields

<i>Differential form</i>	<i>Integral form</i>	<i>Derived from</i>
$\vec{\nabla} \cdot \vec{D} = \rho_v$	$\oint_S \vec{D} \cdot d\vec{S} = \int_V \rho_v dV$	Gauss's Law
$\vec{\nabla} \cdot \vec{B} = 0$	$\oint_S \vec{B} \cdot d\vec{S} = 0$	Nonexistence of magnetic Monopole
$\vec{\nabla} \times \vec{E} = 0$	$\oint_L \vec{E} \cdot d\vec{l} = 0$	Conservativeness of Electrostatic field
$\vec{\nabla} \times \vec{H} = \vec{J}$	$\oint_L \vec{H} \cdot d\vec{l} = \int_S \vec{J} \cdot d\vec{S}$	Ampere's Law

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39